The Patch · La Patch

An on-device, bilingual, n-of-1 method for everyday fatigue, crashes and gut symptoms: method, design rationale and safety rules

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Live: https://v3ktor.org/patch/ · **Source:** https://github.com/tobiedavid/v3ktor.org

Abstract

The Patch is a free, bilingual (English and Canadian French) web application for adults who feel tired, crash during the day, sleep badly, or have loose stools or headaches, and who often run on coffee, energy drinks or alcohol. It runs entirely on the user's phone: there is no account, no server and no tracking. Seventeen questions (eighteen for people who drink) produce a one-week "patch" — what to keep, what to swap and what to add within a stated budget — and the blood test to ask for. After that, a short daily check-in and an optional food log feed three kinds of comparison: habits against the next night's sleep and the next day's energy; families of foods against bad days; and the meals in the hours before each timed crash or symptom. A path orders what to do next: the basics first, then blood work, then the most likely suspect, tested two weeks at a time against the four weeks before, with three days back to normal read at the person's own rate. Thresholds were set and checked by simulation: with no real food effect, about 5% of simulated people see a false food lead at any one look. Hard rules keep it safe: warning signs go to a clinician; pregnancy, people under 18, kidney disease and a history of trouble with eating get no self-guided plain-food test; nobody is told to change a medication; no drink is swapped for another drink. No real-user data exist yet. This document records the method as of 7 November 2026 and is deposited as prior art.

Résumé. La Patch est une application web gratuite et bilingue pour les adultes fatigués, qui ont des crashs dans la journée, dorment mal ou ont des selles molles ou des maux de tête — souvent en carburant au café, aux boissons énergisantes ou à l'alcool. Tout reste dans le téléphone : pas de compte, pas de serveur, pas de pistage. Dix-sept questions (dix-huit pour les personnes qui boivent de l'alcool) donnent une « patch » d'une semaine — quoi garder, quoi remplacer, quoi ajouter selon le budget — et la prise de sang à demander. Ensuite, un court bilan quotidien et un journal alimentaire facultatif permettent de comparer les journées avec et sans chaque habitude ou groupe d'aliments. Un chemin donne l'ordre : la base, puis la prise de sang, puis le suspect le plus probable, testé deux semaines à la fois. Les seuils ont été fixés et vérifiés par simulation; il n'existe encore aucune donnée d'utilisateurs réels. Ce document décrit la méthode au 7 novembre 2026, à titre d'antériorité.

1. Purpose and philosophy

The Patch starts from how people live now. Its first words to the user are that nothing is broken in them and that it isn't laziness; its first job is to patch what is running low — sleep, fuel, protein, iron — without taking anything away. People are asked what they want to change and what they would cut back on, not what they cannot do without. What they don't pick "stays for now", and the app says so plainly if their own numbers later point at it.

Every step offers *I'll do it — remind me* and *I can't — next*; saying "I can't" moves the person to the next-best option instead of stopping. The app was built for one family first, in plain words, in English and Quebec French with French typography. It is education, not medical, pharmacist or dietitian advice, and it is deliberately not an engagement product: no streaks, no ads, no account. All data stay in the phone's browser storage; backups are files the person saves.

2. Inputs

2.1 The questions

Seventeen questions — eighteen for people who drink, who also get one about a usual week — take about three minutes: goals (more energy, sleep better, a calmer gut, lose weight, build muscle, less stress); what keeps the person going (coffee, energy drinks, sugar, beer, wine, spirits, nicotine); what they would cut back on; sleep; stools; when they crash; what they have before noon; medications (among them ADHD stimulants, antidepressants, acid blockers, metformin, birth control, blood thinners, lithium, sleeping or anxiety pills); bleeding; colon cancer in a parent, brother, sister or child; money and time this week; sex, age band, weight and type of work; conditions that change the rules, such as pregnancy, being under 18, kidney disease, or eating vegetarian or vegan; and warning signs.

2.2 The daily check-in

About fifteen seconds: energy (1 to 5), hours of sleep, when the last caffeine was (none, morning, afternoon, evening), drinks (none, 1–2, 3–5, 6 or more) with an optional last-drink time, an optional energy-drink count, hunger while on the weight plan and — for stimulant users only — whether the ADHD pill was taken as usual, late, more than usual, or skipped. A crash and “something was different” (sick, travelling, a period) are one tap each. A day counts in any pattern only once its check-in is finished and the day is over.

2.3 The food log

Two hundred common foods carry per-serving values from Health Canada’s Canadian Nutrient File (2015); where the file is blank, the same or the closest food in USDA SR28 fills the gap and says so. Meals are logged as sittings with a time. “That’s everything I ate” marks a day complete, and only a complete day can count as a day *without* a food.

2.4 Reports

A short crash report (when, how it felt, sleep, caffeine, drinks, what was eaten in the four hours before), symptoms with a time (loose stools, bloating, headache, itch) and blood results (blood count, ferritin with CRP, B12, TSH, vitamin D, the celiac test), each read back in plain words.

3. The one-week patch

From the answers alone, the app writes one week: what to keep, what to swap, what to add within the budget given (from $0), the one thing to do first, and the blood test to ask for, picked from the answers (for example iron stores: ferritin with CRP, so that inflammation cannot hide low iron). Daily needs come from Health Canada’s Dietary Reference Intakes; fibre is set at 14 g per 1,000 kcal. A “fuel tank” shows what the day’s logged food covers against those needs.

4. The path

4.1 Order

Sleep, fuel and deficiencies are treated as the first suspects until they are ruled out. The path has three steps.

- **The basics.** Caffeine not in the evening; no drinks, or drinks finished early; and, on complete food days, at least 60% of the day’s energy need with no single sitting of 1,000 kcal or more that is also over 60% of the need. The basics are met when 10 of the last 14 finished days meet them.
- **Blood work.** Offered when the basics are met and bad days continue — at least 4 bad days among the last 14 finished days, with at least 8 known — or, while bad days continue, after 28 days on the path whether or not the basics are met, or when the person says they can’t do the basics. A bad day is energy at 2 or less, a crash, or a day marked as feeling off. The list is a blood count (CBC), ferritin with CRP, B12, TSH and the celiac test (tTG-IgA with total IgA, taken while still eating gluten). Low or borderline results go to the doctor’s plan first; with no answer after 21 days, the path moves on.

- **The most likely suspect,** one test at a time (section 6).

4.2 Choice

A reminder comes back after 7 days. An offered suspect stays on screen for 14 days, so a lead that wobbles around the bar does not flicker, and a food tested in the last 60 days is not offered again. The person's goals give the reasons for each step; what they said they would cut back on breaks near-ties. Goals never hide a suspect.

4.3 Stimulant users

For people taking an ADHD stimulant, crashes that cluster on pill days are often the medication wearing off. Over the last 28 days, when there are at least 5 pill days with at least 3 crashes, and crashes come at least 25 points more often than on skipped days (or on at least 35% of pill days when there are fewer than 3 skipped days), the path's first card sends the person to their prescriber with the hour the crash hits. The app never says to stop or change a dose, and on pill days a crash is not held against a food.

4.4 Drinks

The app never suggests swapping one alcoholic or energy drink for another: a swap that feels better can end in more of it. When a person's own numbers point at drinks, it says so in their own numbers. The drinks path is: cut back to no more than 2 drinks on any day; for daily drinkers who don't get shaky, "steady nights" instead (no night at 6 or more, the other nights as they are), with a withdrawal stop line; for headaches or itch, the histamine-rich foods with the drinks unchanged; then a pharmacist or doctor. In pregnancy and under 18, the path goes straight to someone who can help.

4.5 Goals and the weight plan

Weight loss only works if the person isn't hungry, with daily needs met and food chosen to keep them full. There are two roads — mostly meat, eggs, cheese and fish, or whole food of every kind with protein at every meal — each with its own rails; a three-week clean start without the usual suspects (measured histamine and lactose foods), then one food group back a week; hunger in the check-in; and an optional target steered by a weekly weigh-in. For stimulant users, who often don't feel hunger, the plan says to eat on the clock. Building muscle sets protein at 1.6 g per kg.

5. Finding suspects

5.1 Families and outcomes

Foods are grouped in families, each compared only with the bad days that make sense for it (Table 1). A loose-stool day is one with loose stools later that day or on waking the next morning.

Family	How a day is "with" it	Compared with
Big dairy servings	one sitting with 12 g of lactose or more, or over 24 g in the day	loose-stool days
Cheese	logged	crash, loose-stool, headache or itch days
Cured meat, canned fish, leftovers	logged	crash, loose-stool, headache or itch days
Eggs	logged	crash, loose-stool, headache or itch days
Shakes, bars, diet drinks, sugar-free sweets	logged	loose-stool and crash days
Energy drinks	logged, or counted in the check-in	crash and loose-stool days
Sweets and sugary drinks	logged	crash and loose-stool days
Apples, pears, mango, watermelon	logged	loose-stool days
High-fibre days	20 g of fibre or more (supplements left out)	loose-stool days
Coffee late in the day	check-in	the next day's crashes

Family	How a day is “with” it	Compared with
3 drinks or more	check-in	the next day’s crashes, headache or itch; loose stools
A big meal; aged or cured foods	crash report only	crashes (3 or more, half of them)

Table 1. Food families and outcomes in v1.2.50. A day is “without” a family only when it is marked complete and has none of it.

5.2 Exposure by dose

Lactose is read by dose: one sitting with 12 g or more, or over 24 g in a day, against complete days under 6 g, because most people who digest lactose poorly tolerate about 12 g at one time (EFSA, 2010). Fibre is read by grams: 20 g or more against complete days under 15 g, with the days in between not counted either way.

5.3 The lead bar

Over the last 60 days, a family becomes a *lead* for an outcome when there are at least 3 bad days with it, at least 3 known days without it, the bad days come at least 30 percentage points more often with it, and a one-sided Fisher exact test (Fisher, 1922) puts a split that lopsided at less than 1 in 50 ($p < 0.02$). The bar was set on 400 simulated people over six weeks, with food logged on 60% of days. With no food effect, 5% got a lead at a given look; a first rule of 2 bad days and 30 points gave 58%, and a 1-in-20 bar gave 13%. With a strong cheese effect (crashes on 50% of cheese days, 15% of other days), cheese was the top lead for 21% by six weeks, more as days add up. Because the app looks every day, the chance of a false lead at some point over six weeks is higher — about 1 in 6 in simulation — which is why a lead is only ever a suspect to test, never a verdict. Once a family has been tested, only the days after the test count for it.

5.4 Foods eaten nearly every day, and calm first

When a food is on nearly every complete day, the counts cannot compare. It is offered as a test — one at a time, after the real leads — when bad days come at least 1 day in 5. When the outcome is on more than half of the last 28 days, everyday foods look guilty and days back cannot tell; the app says so and points to a calmer base first. In simulation, a seven-day bring-back of a food that causes bad days 45% of the time caught it 68% of the time against a calm background (5% bad days) but only 15% against a noisy one (20%), with an innocent food flagged 0.4% of the time in both. Fibre is never offered this way: someone who eats high-fibre every day is not told to test it.

5.5 Drinks by dose

Drinks are read on a ladder: 0–2, 3–5, and 6 or more. The ladder is a lead when bad days rise step by step, the top step is at least 30 points above the bottom, there are at least 3 bad days, and a one-sided Cochran–Armitage trend test (Cochran, 1954; Armitage, 1955) gives $p < 0.02$. It can be read for daily drinkers, who have no day without drinks to compare with.

5.6 The hours before

For each crash or symptom noted with a time, the app looks at the meals in the 4 hours before (6 hours for loose stools or bloating, since fermentation in the large bowel comes later) and compares them with the same clock hours on days the event did not happen, using the same bar. A “without” counts only when every day the window touches is complete. In simulation (300 people, 42 days), 5% got a false hours lead with no food effect; when sweets at lunch caused afternoon crashes (50% of those days against 12%), the hours found it for 28.7% of people by six weeks and the day counts for 3%.

5.7 Days set aside

A day marked “something was different” is left out of the patterns, the tests, the days back, the path and the hours analysis; its drinks and slips still count, and the app shows how many days were set aside. For women the label includes a period: in one survey, 73% of 156 healthy women reported gut symptoms before or during their period (Bernstein et al., 2014).

6. The two-week test

6.1 Design

The person keeps the suspect out for 14 days and everything else as it is. A slip — tapped on the test card, or seen in the log — drops that day out; nothing resets. The 14 days are compared with the four weeks before (or with the days since the last change that helped, leaving out that change's own days back) using a one-sided two-sample Fisher exact test. An earlier design compared with the two weeks before; because a lead tends to appear right after an unlucky stretch, it let an innocent food come out "fewer" about one time in five.

6.2 Verdicts

- **Fewer:** bad days lower during the test, with $p < 0.1$.
- **Too close to call:** at least a third lower but not clear; the person can keep it out two more weeks (once, 28 days in all).
- **No clear change:** this does not rule the food out.
- **Too few days:** fewer than 7 known days before or during. **Nothing to reduce:** no bad days before the test.

In a recalibration on simulated people, an innocent food came out "fewer" in 2–7% of two-week tests (20–22% with the first design), and a real cause that halves crashes in 23–25% — about 40% with the two more weeks. Two weeks only see large effects; "no clear change" is not a clean bill. The design follows the logic of n-of-1 trials (Kravitz & Duan, 2014), simplified to one period off and a short return.

6.3 Days back

After a food test — never after a drinks test, where the two weeks are the answer — the person goes back to normal for 3 days. The trouble "came back" when it shows on at least 2 of the known days back. The app calls that the clearest sign one person can get only when luck alone, at the person's own rate (before and during pooled), would bring it back that often less than 1 time in 10.

6.4 The plain-food test

When loose stools follow high-fibre days, the test is two weeks of plain food: meat, fish and seafood (fresh or frozen), eggs, peeled potatoes or white rice, and mozzarella; vegetarian and vegan versions swap in eggs, firm tofu, lactose-free milk or soy milk, and white pasta. It changes more than fibre, and the app says so: when it helps, foods come back one group a week — cooked vegetables, whole grains, beans and lentils, then raw vegetables and fruit. Potatoes are eaten hot, since part of the starch becomes a fermentable fibre when cooled; psyllium is allowed; the celiac blood test comes first, because the test diet has no gluten. When a smaller family (such as apples and pears) explains the loose stools as well, its test comes first. In simulation, a false fibre lead appeared in 0.4% of 1,200 people with no fibre effect, and the real one in 83% of 400 people with an effect (median day 36).

7. Safety rules

- **Warning signs are not patterns.** Blood in the toilet, black tar-like stool, chest pain or fainting get a plain statement of who to see and how soon (911 for chest pain, fainting or black stool). Thoughts of ending one's life get the 9-8-8 line.
- **Rails.** The plain-food test is not offered in pregnancy, under 18, with kidney disease or a history of trouble with eating; those people are sent to a doctor or dietitian. The weight plan asks about trouble with eating first. Drinks tests are not offered in pregnancy or under 18.
- **Medications.** The app never says to stop, start or change a dose; the prescriber decides.
- **Drinks and brands.** No drink-for-drink swaps; no brand names as recommendations; no energy-drink allowance for teens.
- **Blood work before cutting foods.** The celiac test needs gluten in the diet, and cutting a drink won't fix low iron.

- **Colon cancer in the family.** A parent, brother, sister or child with colon cancer moves screening to age 40: a colonoscopy every 5 years from 40, or 10 years younger than the relative was, when it was found before 60; screening from 40 instead of 50 when it was found at 60 or older (Alberta Health Services; Leddin et al., 2018).

8. Evidence and sources

Every underlined term opens a glossary entry — 72 in all — graded *Solid evidence*, *Likely*, *Worth a try* or *Our reasoning*, with its sources. The 153 sources were opened and checked one by one (PubMed record, guideline or regulator page) in November 2026. Ferritin is read with CRP, after Thurnham et al. (2010) and the WHO (2020). Food values come from the Canadian Nutrient File (Health Canada, 2015) with blanks from USDA SR28, and daily needs from Health Canada's Dietary Reference Intakes.

9. Validation to date

All validation so far is by simulation; no one has yet used the app for real.

Check	Setup	Result
Food lead, no effect	400 simulated people, 6 weeks, food logged on 60% of days	5% false lead at a given look; about 1 in 6 at some point over 6 weeks of daily looks
Food lead, strong effect	crashes on 50% of cheese days vs 15%	top lead for 21% by 6 weeks
The hours before	300 people, 42 days; sweets at lunch → afternoon crashes 50% vs 12%	5% false; found for 28.7% (day counts: 3%)
Two-week verdict	innocent food; real cause that halves crashes	"fewer" in 2–7%; in 23–25% (about 40% with two more weeks)
Background noise	7-day bring-back, trigger causing bad days 45% of the time	caught 68% at a 5% background, 15% at 20%; innocent 0.4%
Fibre lead	1,200 people with no effect; 400 with one; 60 days	false 0.4%; found 83%, median day 36
Pill-day rule	a simulated stimulant user, 25 runs	a food blamed for crashes in 1 run before the rule, 0 after; prescriber first in 22
End to end	simulated people living 70 days in the real app and doing what it says	a fibre-sensitive person offered plain food in 7 of 8 runs (0 before); 4 of 5 finished tests "fewer"

Table 2. Simulation checks behind the thresholds. Simulated people are not people.

10. Scientific status and limits

- The Patch is an educational self-experiment tool, not a diagnostic device. Its outputs are suspects to test and questions to bring to a clinician.
- Two weeks only see large effects; "no clear change" does not rule anything out.
- It depends on logging: the "without" side needs days marked complete, and thin logs slow everything down.
- Many families and outcomes are looked at every day, so false leads happen — about 1 in 6 people over six weeks in simulation. The two-week test is the check.
- Cutting a food changes other things (the plain-food test most of all), and a lead can follow an unlucky stretch; the four-week comparison is there for the second.
- The thresholds are judgement calls set in simulation and must be checked against real logs.
- No pharmacist has reviewed it yet, and its status as software under Health Canada's rules has not been assessed.

11. Data and privacy

All answers, logs and results stay in the phone's browser storage; nothing is sent to a server. That storage is not encrypted, so anyone who can unlock the phone can read it. Meals, symptoms and crash reports are kept for a year, check-ins for 90 days, and backups are files the person saves. Only what the person chooses leaves the phone: a message typed into "Tell Tobie", or a summary they copy to a clinician or an AI chat.

12. Availability

- **Reference implementation:** The Patch v1.2.50 — one HTML file with its offline layer — at https://github.com/tobiedavid/v3ktor.org, commit 2d6628c5964020e9e3b7eb3935dbefb3f2393925, under the GNU Affero General Public License, version 3 or later. Live at https://v3ktor.org/patch/.
- **Food data:** Canadian Nutrient File, Health Canada, 2015 — contains information licensed under the Open Government Licence – Canada; USDA SR28, public domain.
- **This document:** Creative Commons Attribution 4.0 International (CC BY 4.0).

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Colophon

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